

ANEMIA: TELL THEM APART FAST

NUR 122 | Less hemoglobin -> less oxygen delivery -> fatigue, pallor, weakness, and shortness of breath.

PERNICIOUS = ABSORPTION PROBLEM

Lack of intrinsic factor (IF) -> poor B12 absorption -> low B12.
B12 may be in the diet but cannot be absorbed normally.

FOLATE DEFICIENCY = TOO LITTLE B9

Low intake / increased need -> too little folate for RBC formation.
Folate absorption does not require intrinsic factor.

Anemia	What is wrong?	Fast differentiating clues	Typical labs / RBC size	Treatment focus
Iron deficiency	Low iron -> less hemoglobin. Often chronic bleeding, poor intake, or increased need.	Pica + brittle / spoon nails Think GI or menstrual blood loss.	Small RBCs / low MCV Low ferritin and iron; TIBC usually high.	Replace iron; identify and treat the cause of iron loss.
Pernicious (B12 deficiency)	Autoimmune loss of parietal cells / IF -> poor absorption of vitamin B12.	Tingling, numbness, unsteady gait B12 maintains myelin. Glossitis may occur.	Large RBCs / high MCV Low B12; IF antibodies support diagnosis.	Lifelong B12 replacement: IM or prescribed high-dose oral therapy.
Folate deficiency (vitamin B9)	Low folate -> impaired DNA synthesis. Poor diet, alcohol use, pregnancy.	Similar anemia / glossitis, without classic B12 nerve findings. Think low intake / increased need.	Large RBCs / high MCV Low folate. Check B12 too.	Replace folate. Rule out B12 deficiency before giving folate alone.
Hemolytic	RBC destruction is faster than bone marrow replacement.	Jaundice + dark urine RBC breakdown raises bilirubin. Spleen may enlarge.	Often normal MCV. Reticulocytes / indirect bilirubin rise ; haptoglobin often low.	Treat the cause of hemolysis; monitor worsening anemia.
Sickle cell (hemolytic type)	Inherited HbS -> sickling -> hemolysis and blocked blood flow .	Severe pain / vaso-occlusion Triggers: dehydration, infection, low oxygen.	HbS on hemoglobin testing. Usually normal MCV; reticulocytes often high.	Pain control; correct dehydration; O2 if hypoxemic. Hydroxyurea reduces crises.
Acute blood loss	Bleeding -> loss of circulating volume and RBCs.	Bleeding + fast pulse / low BP Think surgery, injury, GI bleed.	Usually normal MCV. Hgb / Hct may initially look normal.	Control bleeding; support circulation. Fluids / blood as ordered.

MCV shortcut: Small <80 fL = think iron | Normal 80-100 = blood loss / hemolysis | Large >100 = B12 / folate. Patterns are clues, not proof.

Megaloblastic = impaired DNA synthesis, usually from B12 or folate deficiency; not a separate missing nutrient. **B12 trap:** Not all B12 deficiency is pernicious; poor intake or other malabsorption can also cause it. No nerve symptoms does not rule B12 deficiency out. Folate alone can mask ongoing B12 nerve damage.

Priority: Bleeding with instability, chest pain, or severe dyspnea needs urgent action. Sickle cell + fever / chest pain / breathing trouble: consider acute chest syndrome.

Course: Anemia CMcCain, slides 5-9, 11, 14-20, 22-25. Clarification: IF helps absorb B12, not iron (slide 14); other GI malabsorption is not automatically pernicious (slide 16).

References: NHLBI: B12-deficiency anemia; Merck: B12 deficiency; Merck: evaluation of anemia.